

C++ <mutex> Header - Components Overview

Key Components in <mutex>

Component	Description
std::mutex	Basic mutex for exclusive locking (not recursive).
std::recursive_mutex	Allows the same thread to lock multiple times.
std::timed_mutex	Supports timeout operations with try_lock_for / until.
std::recursive_timed_mutex	Recursive + timed locking capabilities.
std::lock_guard<T>	RAII wrapper that locks on creation, unlocks on destruction.
std::unique_lock<T>	More flexible lock object with manual lock/unlock support.
std::once_flag	Used with call_once to guarantee single execution.
std::call_once	Executes a callable once, thread-safe.
std::try_lock()	Attempts to lock multiple mutexes without deadlock.
std::lock()	Locks multiple mutexes atomically.
std::adopt_lock / defer_lock / try_to_lock	Flag types to control lock behavior.